Componentwise kNN-Isomap

From neighborhood graphs to constrained manifold embeddings

Mathematical lecture · ProximityGraphs-Isomap · 19 October 2026

Graph theory · Geodesic distances · Classical MDS · Orthogonal alignment · Topology

Data: versioned EXP-010 and EXP-011 full runs, seed 20261018. **New algorithmic claims:** clearly marked as proposed.

What you should understand at the end

1. Why Isomap estimates **geodesic distances** using graph shortest paths.
2. How the symmetrized (k)-nearest-neighbor graph determines the partition.
3. How classical MDS follows from a centered Gram matrix.
4. Why independent embeddings are not positioned relative to one another.
5. Why **rigid motions preserve computed within-component distances**.
6. How to quantify and constrain **intercomponent placement**.
7. What our EXP-011 implementation actually does—and what remains to be developed.

Problem setup: multiple manifolds

Let

$$X = \{x_1, \ldots, x_n\} \subset \mathbb{R}^D, \qquad Y = \{y_1, \ldots, y_n\} \subset \mathbb{R}^d, \quad d \ll D.$$

Suppose observations come from a disjoint union of supports

$$M = M_1 \sqcup M_2 \sqcup \cdots \sqcup M_K.$$

We want to preserve the **intrinsic geometry inside** each (M_c), but **cross-manifold intrinsic geodesic distances are not finite**.

Between distinct components we may still measure **ambient** distances, (|x_i-x_j|_2). These objectives are fundamentally different.

Why ordinary global Isomap is troublesome

Classical Isomap: neighbor graph (\rightarrow) shortest-path matrix (\rightarrow) classical multidimensional scaling (MDS).

When (G) is disconnected, some (d_G(i,j)=+\infty). A single standard MDS matrix cannot use these entries directly.

Some implementations **repair** a disconnected graph by inserting bridging edges. Then global geodesics use connections **not present in the original neighbor graph**.

Alternative: decompose first, unfold each graph component independently, then solve the *placement problem* explicitly.

This idea is part of existing multi-manifold-learning literature; it is **not novel by itself**.

The algorithm, in one picture

$$X \xrightarrow[\text{distance}]{\text{kNN graph}} C_1, \ldots, C_K \xrightarrow[\text{per component}]{\text{shortest paths}} D_1, \ldots, D_K \xrightarrow[\text{per component}]{\text{MDS}} Z_1, \ldots, Z_K \xrightarrow{\text{rigid placement}} Y$$

Separate questions:

- **Discovery:** which observations should belong together?
- **Unfolding:** how should each discovered piece be represented in (\mathbb R^d)?
- **Alignment:** where should each piece be placed relative to the others?

We will derive every arrow.

Step 1 — Nearest-neighbor definition

For each (i), let (N_k(i)) be the indices of the (k) nearest *other* observations under Euclidean ambient distance.

Using the **symmetrized union kNN graph**, define

$$\{i, j\} \in E_k \quad \Longleftrightarrow \quad j \in N_k(i) \text{ or } i \in N_k(j).$$

If an edge exists, assign its Euclidean length:

$$w_{ij} = \|x_i - x_j\|_2.$$

The union rule is distinct from **mutual kNN**, which requires *both* neighbor relations. These choices have different connectivity behavior.

Why the graph is weighted

Consider three observations (a,b,c). A path (a\rightarrow b\rightarrow c) has cost

$$L(a,b,c) = \|x_a - x_b\|_2 + \|x_b - x_c\|_2.$$

The graph-shortest distance is

$$d_G(i,j) = \min_{i=v_0,\ldots,v_q=j} \sum_{\ell=0}^{q-1} \|x_{v_\ell} - x_{v_{\ell+1}}\|_2.$$

When the neighborhood edges follow the manifold, graph distances can approximate surface/geodesic distances that are **longer than** direct ambient chords.

But: edges that shortcut across nearby folds spoil the approximation.

Step 2 — Connected components

The undirected graph (G_k=(V,E_k,w)) induces a partition

$$V = C_1(k) \sqcup \cdots \sqcup C_{K(k)}(k).$$

We may compute this partition by depth-first search or a sparse connected-component routine.

For a fixed dataset and deterministic tie-breaking,

$$E_k \subseteq E_{k+1} \quad \Longrightarrow \quad K(k+1) \leq K(k).$$

As (k) increases, graph components can merge but cannot split.

This statement concerns graph connectivity—not whether the resulting groups equal the true generating manifolds.

What can go wrong when choosing (k)?

Too small: locally sparse manifolds fragment, creating too many components and infinite paths between fragments.

Too large: points on different supports can become nearest neighbors, merging manifolds into one connected graph.

Intermediate (k): within-manifold connectivity may coexist with no inter-manifold bridges.

This interval need not exist when manifolds nearly intersect, density varies strongly, or high-dimensional nearest neighbors become unreliable.

EXP-013 proposal: map the entire (k)-dependent component and distortion landscape, instead of picking one attractive example.

An exact finite-sample recovery condition

For true generating groups (X_1,\ldots,X_K), a symmetrized kNN partition equals them if:

1. Each induced within-group kNN subgraph is **connected**.
2. Every point's chosen (k) neighbors are **inside its own true group**.

Under these two conditions, the graph has exactly those (K) components.

This is a sufficient condition, not an unsupervised test. It uses known generating groups to state a guarantee; real data do not supply these groups.

Contrast radius graphs, for which a sufficient recovery interval follows from the largest within-group MST edge and smallest cross-group distance.

A crucial EXP-011 implementation detail

The current experiment calls its method **componentwise kNN Isomap**, but its stages are:

1. Find components using a **radius graph** (rings, cylinder, torus), or a **supplied-(K) MST cut** (Swiss/S-curve pairs).
2. **Rebuild a symmetrized 10-NN graph inside each detected component**.
3. Compute its graph shortest paths.
4. Apply MDS and place embeddings in separated strips.

Therefore this run does not test unsupervised kNN component discovery.

A new specialized experiment should explicitly vary both detection (k_{\rm detect}) and unfolding (k_{\rm embed}).

Two neighborhood parameters—not necessarily one

For a fully kNN-driven pipeline, it is useful to distinguish

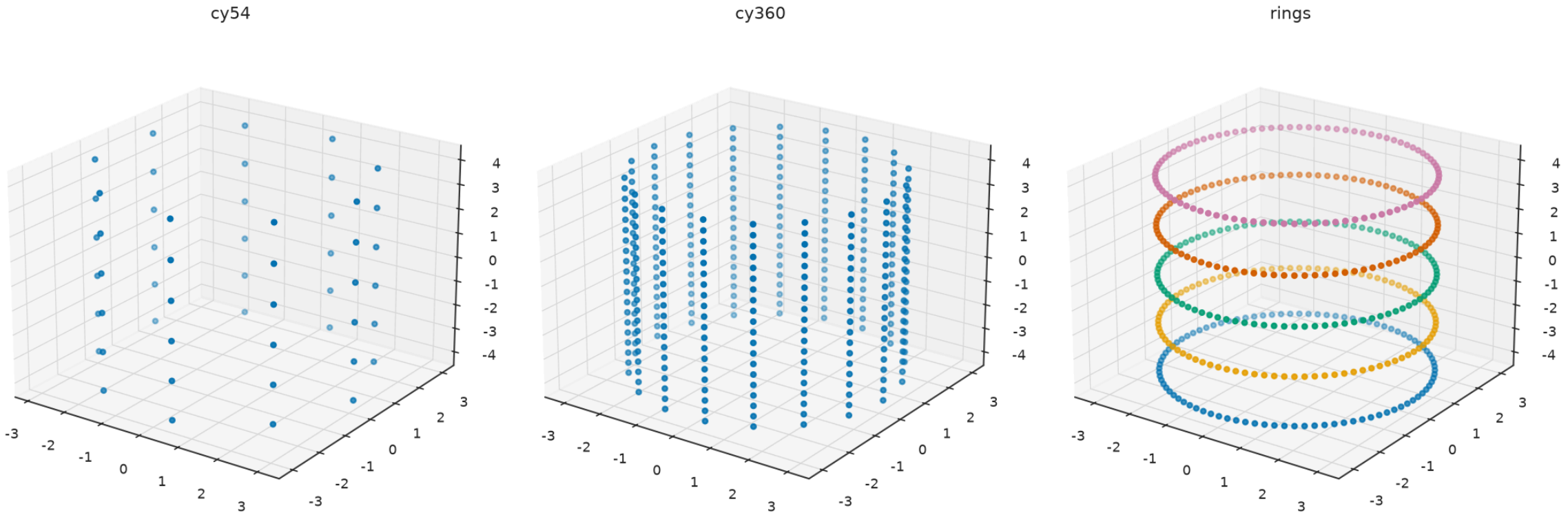
$$k_{\mathrm{detect}} \quad \text{and} \quad k_{\mathrm{embed}}.$$

- (k_{\rm detect}) determines candidate manifold components.
- (k_{\rm embed}) determines shortest-path geometry *inside* each candidate component.

We can set them equal for a simple algorithm, but the two choices answer different geometric questions.

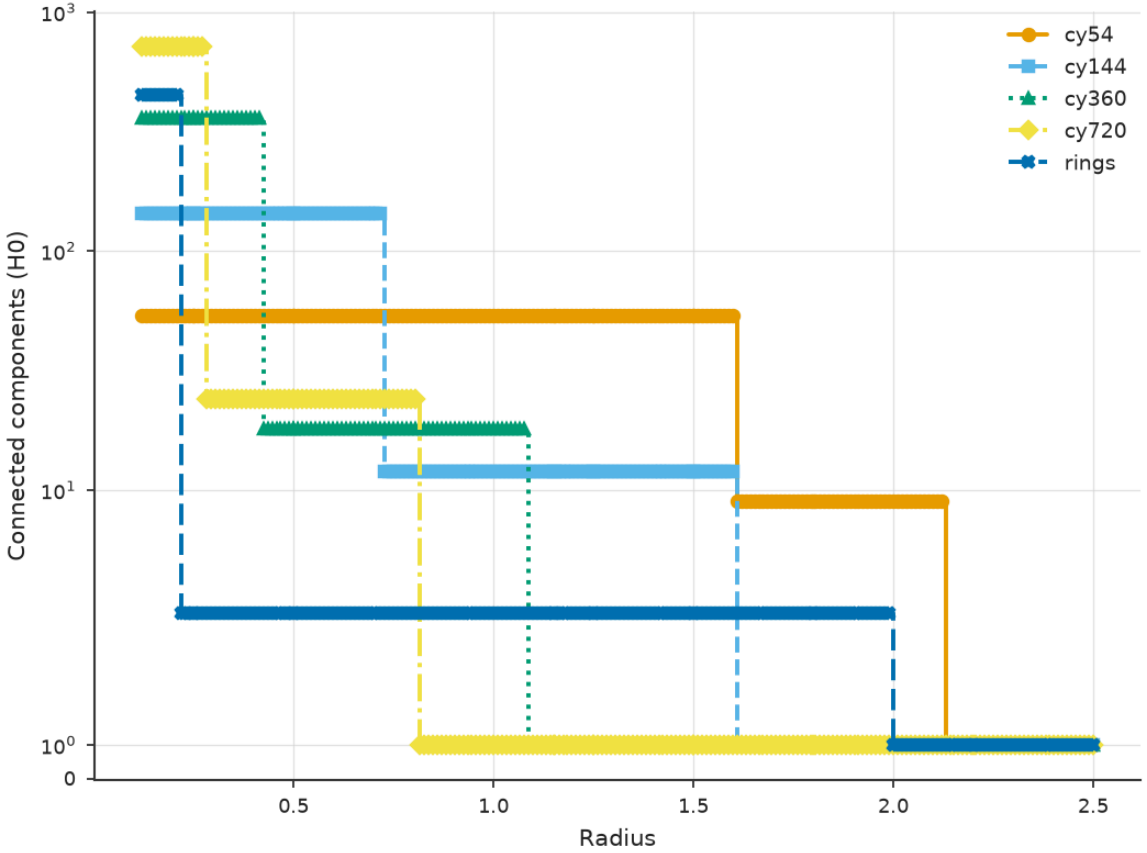
Using an induced subgraph (G_{k_{\rm detect}}[C_c]) is also different from **recomputing** (k_{\rm embed})-NN within the selected component; specify which is used.

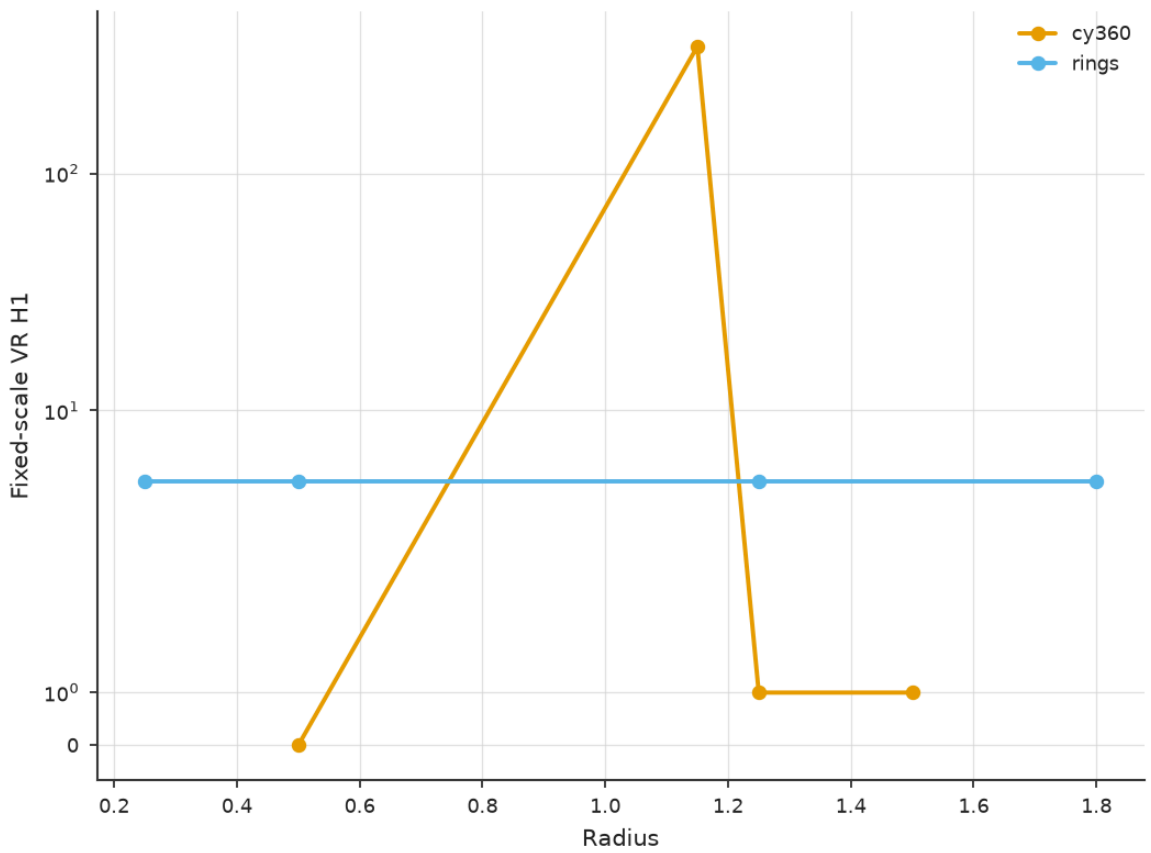
EXP-010 — Why rings and a cylinder are different



Executed EXP-010: the sampled cylinder is one ideal connected surface; the five rings are distinct connected supports.

EXP-010 — Radius affects connectivity and H1





Observed result: β_0 counts graph components, whereas β_1 counts independent VR loops. Smaller radius does not convert one cylinder into five genuine supports.

Step 3 — Shortest paths within a component

Let (C_c) have (n_c) observations and a **connected** local kNN graph (G_c).

Construct the matrix

$$D_c = (d_{ij}^{G_c})_{i,j \in C_c} \in \mathbb{R}^{n_c \times n_c}.$$

The graph shortest-path distance is finite within a connected (C_c), and the diagonal is zero.

Important assumption: we need local enough edges to avoid shortcuts yet enough edges to connect the sampled manifold. Neither property is guaranteed by choosing (k) alone.

Why geodesics matter: the unrolling intuition

A Swiss roll is a bent two-dimensional sheet. Nearby points in (\mathbb R^3) can be far apart **along the sheet**.

- PCA tries to preserve variance in a single linear plane.
- Direct Euclidean distance MDS would attempt to preserve ambient chords.
- Isomap replaces ambient chords with **graph approximations to surface geodesics**.

When those graph distances approximate true intrinsic geodesics, MDS can recover a useful intrinsic coordinate representation.

A wrong graph cannot be repaired by MDS.

Step 4 — Classical MDS: begin with coordinates

Let (z_1,\ldots,z_m\in\mathbb R^d) be *unknown* centered coordinates,

$$\sum_{i=1}^{m} z_i = 0.$$

Define the Gram matrix

$$B_{ij} = \langle z_i, z_j \rangle, \qquad B = ZZ^\top.$$

The squared Euclidean distance satisfies

$$D_{ij}^2 = \|z_i - z_j\|_2^2 = B_{ii} + B_{jj} - 2B_{ij}.$$

Goal: recover a good coordinate matrix (Z) from distances (D), without knowing the original coordinates.

The centering matrix and why it works

Define

$$H = I - \frac{1}{m}\mathbf{1}\mathbf{1}^\top.$$

Then

$$H^\top = H, \qquad H^2 = H, \qquad H\mathbf{1} = 0.$$

For an arbitrary coordinate matrix (W), (HW) subtracts the row mean.

Multiplying the squared-distance identity on both sides by (H), the diagonal terms vanish because they involve the all-ones vector:

$$HD^{\circ 2}H = -2HBH.$$

Derivation: the double-centered Gram matrix

For centered (Z), we have (HZ=Z), hence (HBH=B). Therefore

$$\boxed{B = -\frac{1}{2} H D^{\circ 2} H.}$$

This is the central mathematical identity behind classical MDS.

Starting from graph-geodesic distances (\widehat D_c), form

$$\widehat{B}_c = -\frac{1}{2} H_c \widehat{D}_c^{\circ 2} H_c.$$

Because graph distances need not be exactly Euclidean in (d) dimensions, (\widehat B_c) may contain negative eigenvalues.

MDS as a spectral approximation

Diagonalize the symmetric matrix

$$\widehat{B}_c = U\Lambda U^\top, \qquad \lambda_1 \geq \cdots \geq \lambda_{n_c}.$$

Keep up to (d) leading **positive** eigenvalues:

$$\boxed{Z_c = U_{[:,1:d]}\,\mathrm{diag}\big(\sqrt{\max(\lambda_1, 0)}, \ldots, \sqrt{\max(\lambda_d, 0)}\big).}$$

If fewer than (d) eigenvalues are positive, pad with zero coordinates.

This is the best **positive-semidefinite rank-(d)** approximation of (\widehat B_c) in Frobenius norm. It does **not** directly minimize the raw unsquared pairwise-distance stress.

What exactly does classical MDS optimize?

Writing (Z\in\mathbb R^{m\times d}),

$$\min_{Z} \|\widehat{B}_c - ZZ^\top\|_F^2$$

is solved by the truncated positive eigendecomposition.

This is **strain minimization** in a centered Gram-matrix representation.

A separate objective is **stress**:

$$\sum_{i<j} \left(\|z_i - z_j\|_2 - \widehat{d}_{ij}\right)^2.$$

They are not the same optimization problem. When presenting Isomap, do not claim its classical MDS step globally minimizes raw distance stress.

Step 5 — Apply MDS independently

For each candidate component (C_c):

$$G_c \;\to\; D_c \;\to\; B_c = -\tfrac{1}{2} H_c D_c^{\circ 2} H_c \;\to\; Z_c.$$

This gives (Z_c\in\mathbb R^{n_c\times d}).

There is no distance matrix linking distinct (C_c) via source-manifold geodesics. Their independent MDS solutions thus have no common translation or orientation.

This is why stacking all (Z_c) at the origin—even when each component individually unfolds well—can produce overlapping geometry.

A freedom we can exploit: rigid motions

Choose an orthogonal matrix (Q_c\in O(d)), (Q_c^\top Q_c=I), and a translation (t_c\in\mathbb R^d).

Set

$$\boxed{Y_c = Z_c Q_c + \mathbf{1} t_c^\top .}$$

(Q_c) allows rotations **and reflections**, while (t_c) repositions the cloud.

A **single global rigid motion** cannot change cross-component relationships, but **independent rigid motions** can.

All of this happens *after* the independent Isomap embedding is computed.

Proof: the internal distances cannot change

For (i,j\in C_c),

$$y_i - y_j = (z_i - z_j)Q_c.$$

Consequently,

$$\begin{aligned}\|y_i - y_j\|_2^2 &= (z_i - z_j)Q_cQ_c^\top(z_i - z_j)^\top \\ &= (z_i - z_j)(z_i - z_j)^\top \\ &= \|z_i - z_j\|_2^2.\end{aligned}$$

Therefore rigid placement preserves every pairwise Euclidean distance of the computed embedding exactly.

It does **not** prove Isomap recovered the true input manifold metric.

What rigid motions do NOT preserve

For (i\in C_c), (j\in C_\ell), (c\ne\ell),

$$\|y_i - y_j\|_2 = \|z_i Q_c + t_c - z_j Q_\ell - t_\ell\|_2.$$

This distance depends on **relative rotations and translations**.

Therefore independent positioning can:

- destroy ambient cross-component distance fidelity;
- introduce or prevent component overlap;
- alter the Vietoris–Rips filtration of the **union** of projected components.

These are the variables in the proposed placement optimization.

Step 6 — Defining geometric separation

A graph partition only separates *vertices in the graph*, not necessarily the projected shapes.

A strong geometric requirement for (c\ne\ell) is

$$\boxed{\operatorname{dist}\big(\operatorname{conv}(Y_c), \operatorname{conv}(Y_\ell)\big) \geq \mu > 0.}$$

Here

$$\operatorname{dist}(A, B) = \inf_{a \in A,\ b \in B} \|a - b\|_2.$$

Using convex hulls avoids the error of calling two point clouds 'separated' merely because no source graph edge connects them.

Constructive guarantee: disjoint strips

For each centered (Z_c), choose a translation such that the first coordinates lie inside a strip (I_c=[a_c,b_c]).

Order the components so that

$$a_{c+1} - b_c \geq \mu.$$

Since every convex combination retains a first coordinate inside the same interval (I_c), each convex hull lies within its strip.

For points (p\in\operatorname{conv}(Y_c)), (q\in\operatorname{conv}(Y_\ell)), the horizontal difference is at least (\mu), and

$$\|p - q\|_2 \geq |p_1 - q_1| \geq \mu.$$

Proved: strip placement guarantees clearance; **not proved:** that it minimizes cross-component distance distortion.

EXP-011: separation may be expensive

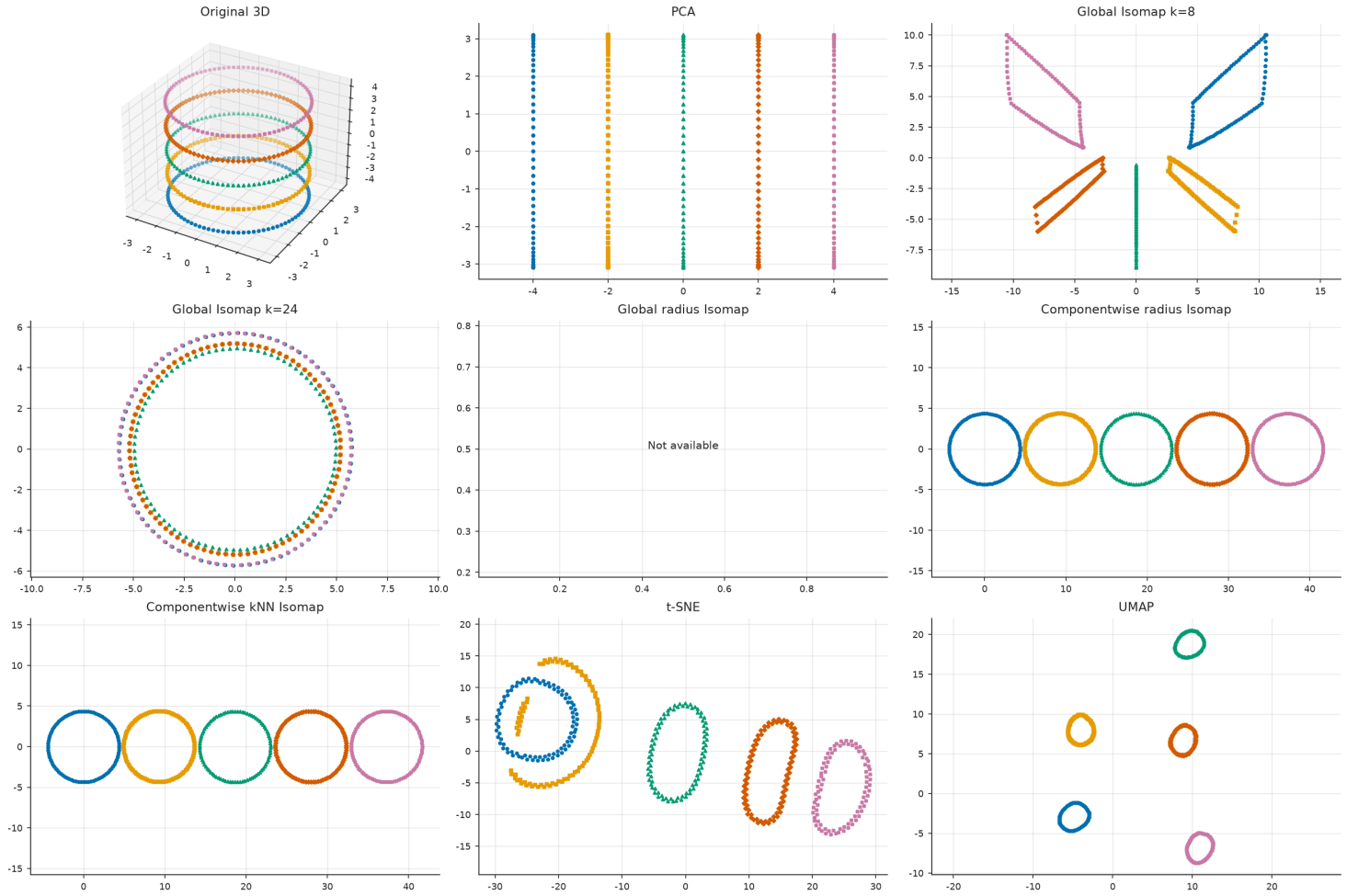
For the five rings in the *executed* EXP-011 run:

Method	Intrinsic stress ↓	Centroid stress ↓	Intersecting hull pairs ↓
PCA	0.463	**0.414**	0
Global Isomap (k=24)	0.250	0.996	10
Componentwise kNN Isomap	**0.245**	2.311	**0**
UMAP	0.253	6.263	0

Tradeoff: the componentwise method unfolds/separates the rings here, but its simple strip placement distorts the component centroids.

Run: `20261019T061001Z-full`, one deterministic seed. No significance claim.

Real EXP-011 — Five rings across the methods



The 2D gallery corresponds to the GitHub-published run; each panel uses the same 450 observed source points.

Step 7 — Centroid-distance preservation

Let (\bar x_c) be the source centroid and (\bar y_c) the projected centroid. Consider

$$\mathcal{L}_{\mathrm{cent}}(Y) = \sum_{c<\ell} w_{c\ell}\big(\|\bar{y}_c - \bar{y}_\ell\|_2 - \|\bar{x}_c - \bar{x}_\ell\|_2\big)^2.$$

A normalized version divides by
(\sum_{c<\ell}w_{c\ell}|\bar x_c-\bar x_\ell|_2^2).

But centroid distances only describe a **coarse arrangement**; they do not determine component orientations or preserve every intercomponent point distance.

A useful limitation: centroids do not fix orientation

If each independent embedding is centered, (\bar z_c=0), then

$$\bar{y}_c = \bar{z}_c Q_c + t_c = t_c.$$

Thus

$$\mathcal{L}_{\mathrm{cent}}(Y) = \sum_{c<\ell} w_{c\ell}\big(\|t_c - t_\ell\| - \|\bar{x}_c - \bar{x}_\ell\|\big)^2.$$

The centroid loss is independent of every (Q_c)!

Therefore centroid stress alone cannot select optimal orientations. A meaningful placement algorithm needs separation constraints and possibly landmark or all-pair cross-component terms.

Initial centroid placement via classical MDS

Take the (K\times K) matrix of original centroid distances,

$$D^{\mathrm{cent}}_{c\ell} = \|\bar{x}_c - \bar{x}_\ell\|_2.$$

Apply classical MDS to (D^{\rm cent}) in dimension (d). The resulting (T\in\mathbb R^{K\times d}) supplies initial translations (t_c).

Because (K) original centroids may have affine dimension above (d), exact reproduction of **all** centroid distances is generally impossible.

Then optimize the component orientations and translations **subject to positive geometric clearance**.

Stronger loss: intercomponent landmarks

Pick representative landmarks (L_c\subset C_c). Define

$$\mathcal{L}_{\text{cross}}(Q,t) = \sum_{c<\ell} \sum_{i\in L_c} \sum_{j\in L_\ell} w_{ij}(\|z_i Q_c + t_c - z_j Q_\ell - t_\ell\| - \|x_i - x_j\|)^2.$$

This loss **does** depend on orientation (Q_c).

It matches **ambient cross-component distances**—not intrinsic geodesics, which are disconnected across different manifold supports.

Landmarks control cost, but must be selected without leaking evaluation targets.

The proposed constrained optimization

The intended next method solves

$$\begin{aligned} \min_{Q_c, t_c} \quad & \mathcal{L}_{\text{cross}}(Q, t) \\ \text{subject to} \quad & Q_c^\top Q_c = I_d, \quad \forall c, \\ & \operatorname{dist}\big(\operatorname{conv}(Y_c), \operatorname{conv}(Y_\ell)\big) \geq \mu, \quad c \neq \ell. \end{aligned}$$

Implemented in EXP-011: independent Isomap and *feasible strip placement*.

Not implemented in EXP-011: global optimization of this constrained objective. Do not describe the strip layout as an optimizer or an original algorithmic invention.

Separation–distortion frontier

For a chosen target clearance (\mu), define the ideal value

$$V(\mu) = \inf_{\{Q_c,t_c\}\,:\,\operatorname{dist}(\operatorname{conv} Y_c,\operatorname{conv} Y_\ell)\ge\mu} \mathcal{L}_{\text{cross}}(Q,t).$$

If (0\le\mu_1\le\mu_2), then the feasible set for (\mu_2) is contained in that for (\mu_1). Hence

$$\boxed{V(\mu_1) \le V(\mu_2).}$$

This monotonicity is **elementary**. A publishable research problem would need stronger bounds, an effective optimizer, or robust comparative evidence.

Step 8 — Evaluating internal distances

When synthetic ground-truth within-manifold geodesics (g_{ij}) are available, use

$$S_{\mathrm{in}} = \sqrt{\frac{\sum_c \sum_{i<j\in C_c} (\|\tilde{y}_i - \tilde{y}_j\|_2 - g_{ij})^2}{\sum_c \sum_{i<j\in C_c} g_{ij}^2}}.$$

To handle arbitrary scales across embeddings, EXP-011 first applies **a common ambient-within-distance RMS scale factor** (s(Y)) to obtain (\tilde Y=s(Y)Y).

Interpretation: this evaluates intrinsic distances in a specified synthetic geometry. It does **not** replace graph-connectivity diagnostics.

Step 9 — Evaluating intercomponent placement

The distinct manifolds are disconnected; we compare their projected distances with finite source **ambient** distances:

$$S_{\text{out}} = \sqrt{\frac{\sum_{c<\ell} \sum_{i \in C_c, j \in C_\ell} (\|\tilde{y}_i - \tilde{y}_j\| - \|x_i - x_j\|)^2}{\sum_{c<\ell} \sum_{i \in C_c, j \in C_\ell} \|x_i - x_j\|^2}}.$$

Complement this with centroid stress and **convex-hull intersection counts**.

If a method has low (S_{\rm in}) but high (S_{\rm out}), it has unfolded the pieces while losing their relative ambient placement.

Step 10 — Trustworthiness of neighborhoods

Let (U_i(k)) be points that enter the (k)-nearest-neighbor set of (y_i) but were not among the original (k) nearest neighbors of (x_i).

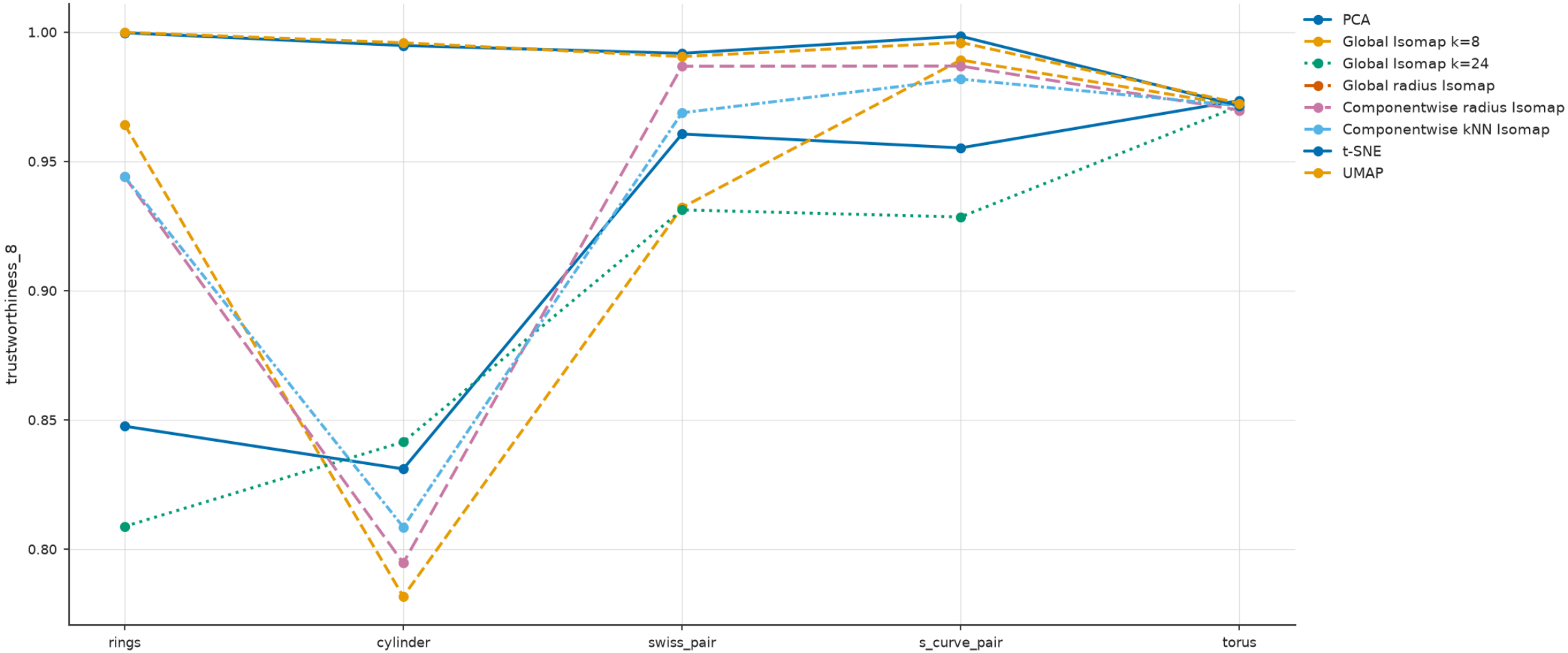
With (r_i(j)) the original-space neighbor rank,

$$T(k) = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^{n} \sum_{j \in U_i(k)} (r_i(j) - k).$$

(T(k)\approx1) means **few false neighbors introduced by projection**.

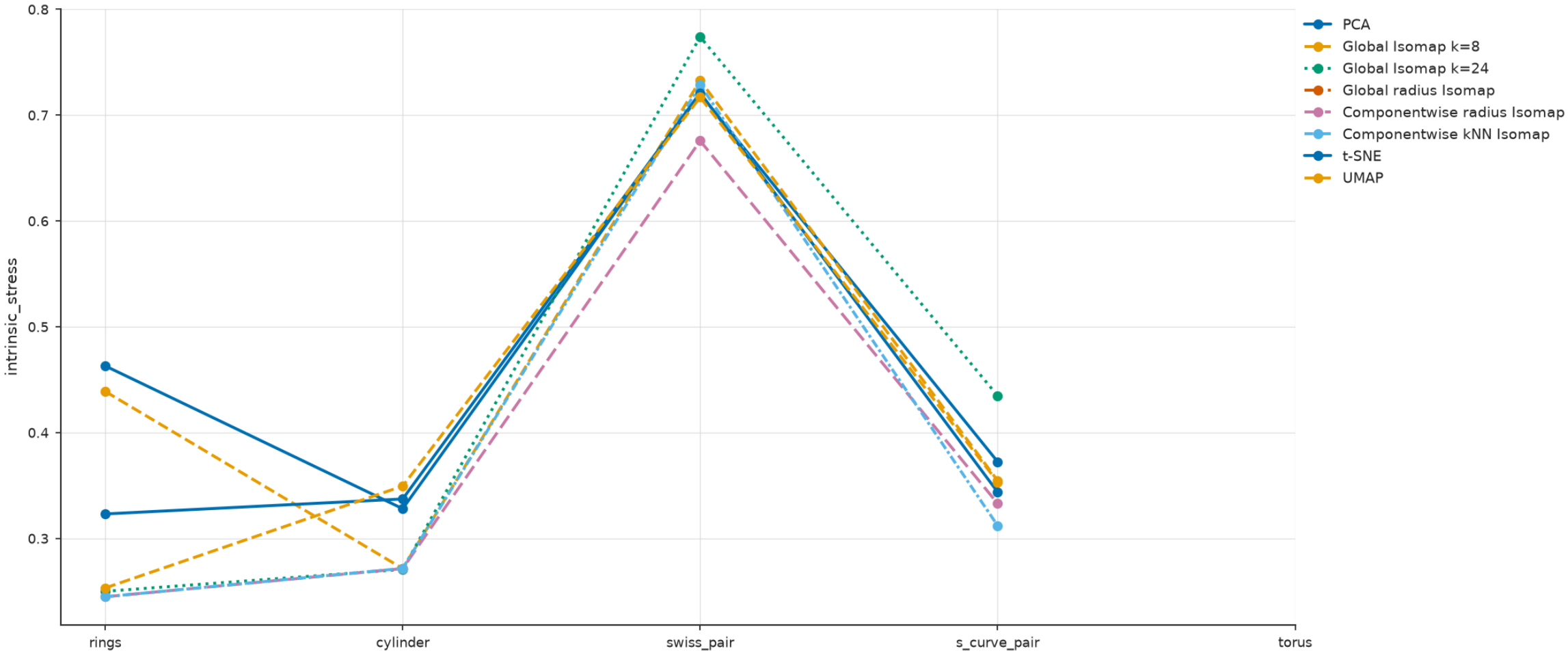
This is useful for t-SNE and UMAP, whose objectives primarily favor neighborhood structure. High trustworthiness does **not** imply correctly preserved centroids or nonoverlapping hulls.

Real EXP-011 — Local trustworthiness



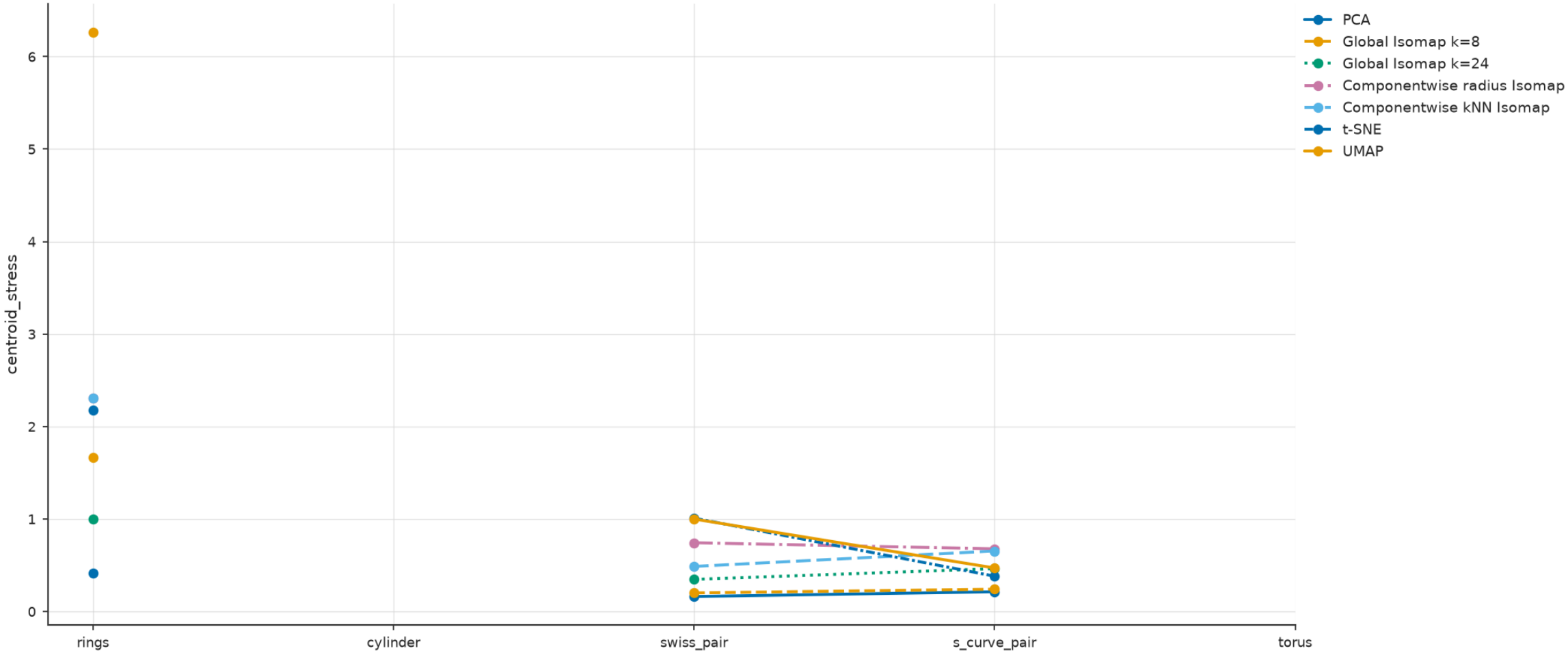
One completed run: t-SNE and UMAP achieve high neighborhood fidelity, even when global metric placement differs.

Real EXP-011 — Intrinsic-distance error



The torus has no reported analytic surface-geodesic stress in this protocol; it must not be treated as zero.

Real EXP-011 — Centroid placement error



For a single connected component centroid-distance stress is undefined; no such number is claimed.

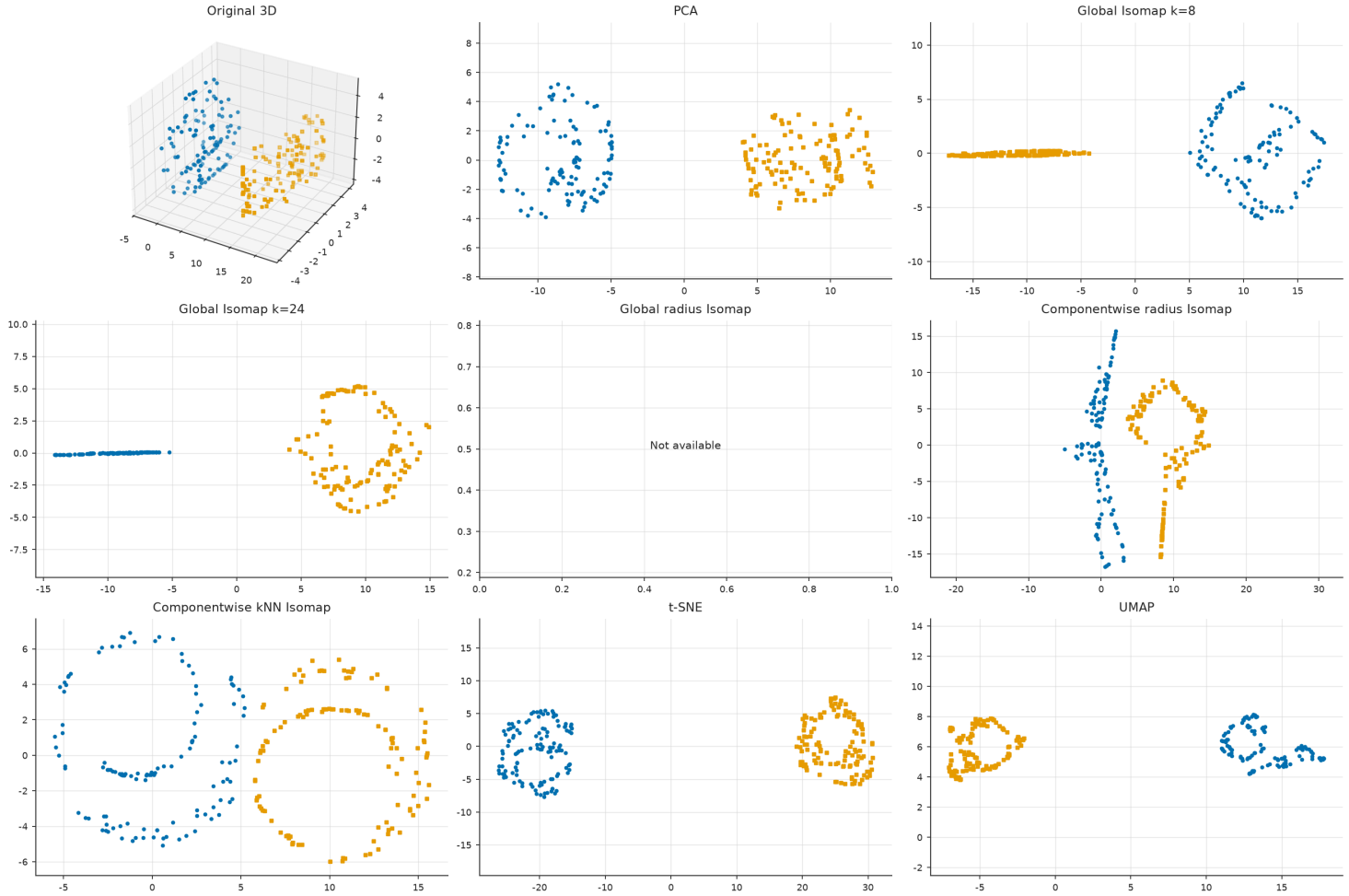
The measured methods target different properties

EXP-011, single run:

- On **five rings**, componentwise kNN-Isomap: intrinsic stress (0.245); trustworthiness (0.944).
- On **five rings**, t-SNE: intrinsic stress (0.323); trustworthiness (\approx0.9997).
- On **five rings**, UMAP: intrinsic stress (0.253); trustworthiness (\approx0.9998).
- On the **S-curve pair**, componentwise kNN-Isomap has lower intrinsic stress (0.312) than PCA (0.372) and global Isomap (k=8), (0.355).
- On the **Swiss-roll pair**, componentwise kNN-Isomap stress (0.729) is **not** a convincing improvement over PCA (0.720).

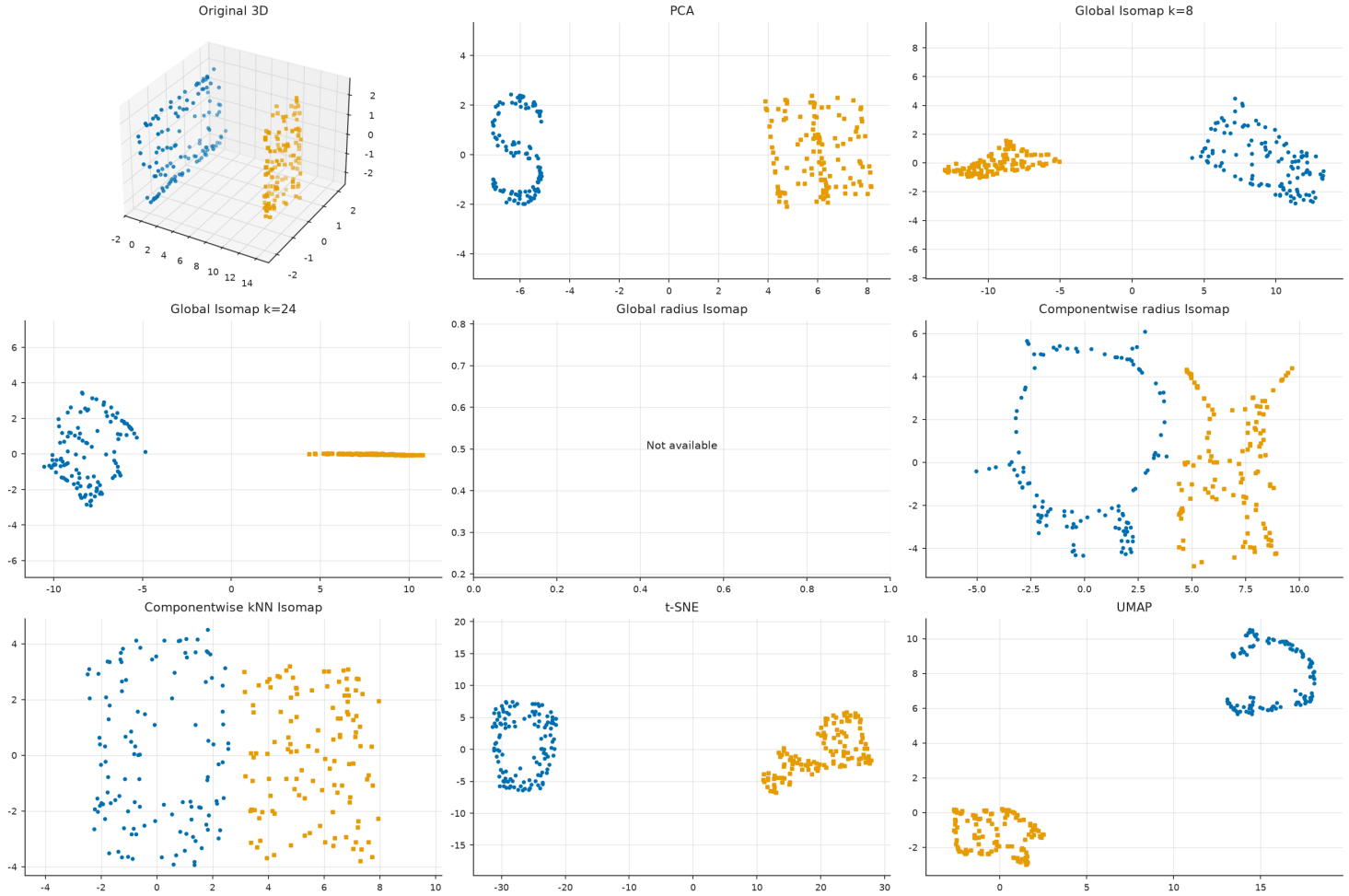
No universal winner; each metric and geometry answers a different question.

Real EXP-011 — Rotated Swiss-roll pair



A difficult negative example: the componentwise kNN graph does not automatically give low intrinsic error on these rotated random samples.

Real EXP-011 — S-curve pair



An example where the componentwise kNN method has relatively lower within-manifold stress in this run.

Does a good 2D embedding preserve topology?

Not necessarily.

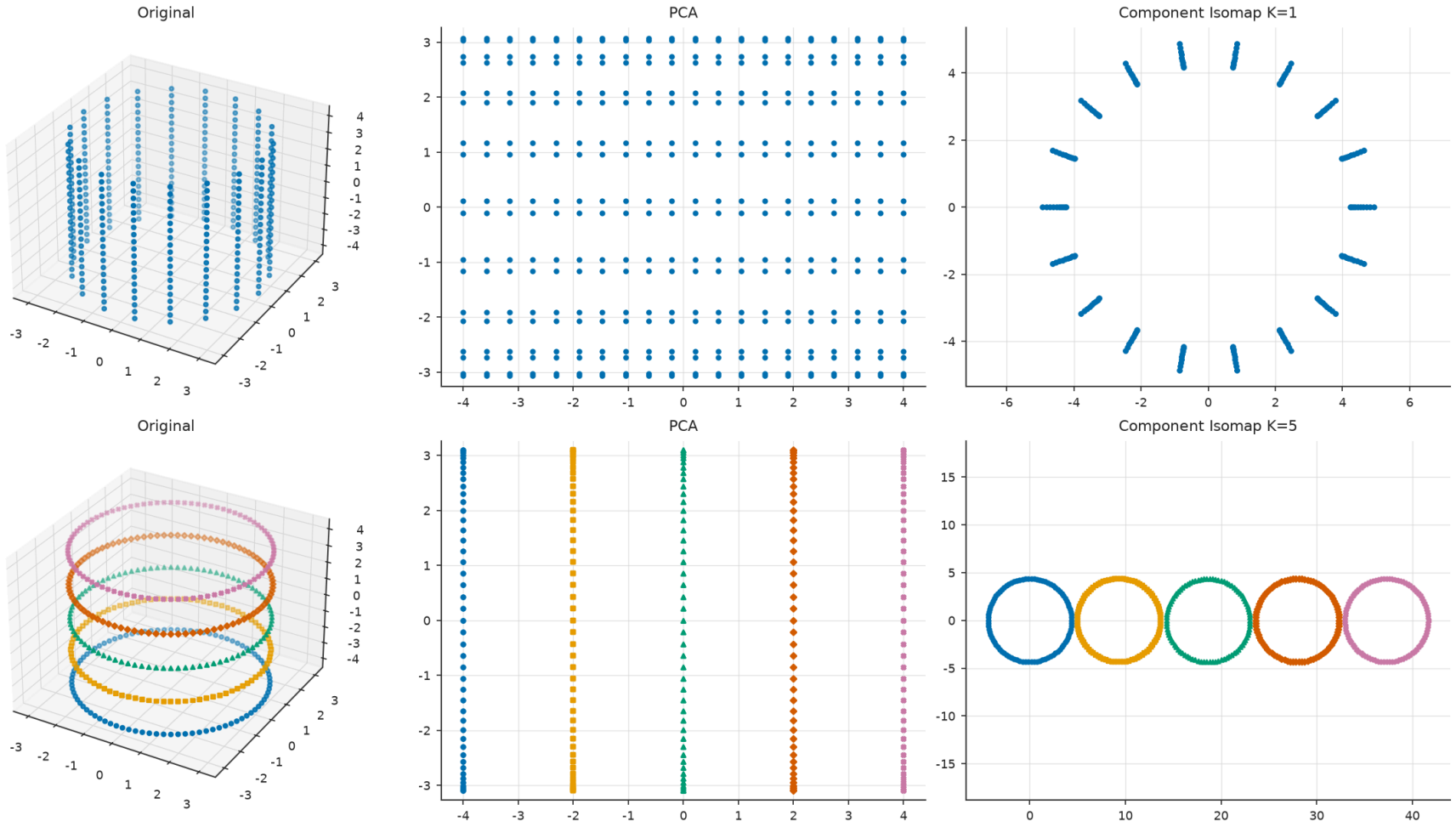
For a point cloud (X), the Vietoris–Rips complex is

$$\mathrm{VR}_r(X) = \{\sigma \subseteq X : \max_{i,j\in\sigma} \|x_i - x_j\| \leq r\}.$$

- (\beta_0(r)): connected components.
- (\beta_1(r)): independent 1-cycles *after filling graph triangles*.
- The VR persistence of (Z_c) is unchanged under its **rigid motion**.
- But the ambient-cloud VR persistence of (X_c) can differ substantially from that of **its Isomap embedding** (Z_c).

A torus surface also cannot be faithfully flattened into the plane without topological distortion.

Real EXP-010 — Original, PCA and componentwise Isomap



This control distinguishes a continuous cylinder from five disconnected circular supports.

Computational complexity

Let (n=\sum_c n_c) and (m_c=|E(G_c)|).

- Brute-force neighbor distances: (O(n^2D)) time, potentially (O(n^2)) memory (algorithm/data dependent).
- Build symmetrized kNN edges: typically (O(nk)) graph storage after neighbor search.
- Repeated shortest paths (one source per vertex): roughly (O(\sum_c n_c(m_c+n_c)\log n_c)) with binary-heap Dijkstra, depending on implementation.
- Full eigendecomposition for each component: (O(\sum_c n_c^3)) time and (O(\sum_c n_c^2)) memory.
- Computing all cross-component pair distortions: (O(\sum_{c<\ell}n_cn_\ell d)).

Why componentwise helps: (\sum_c n_c^3\le n^3), sometimes much smaller; cost of graph discovery and placement remains.

The experimental design for EXP-013 (proposed)

Separate the two stages rather than mixing effects:

$$k_{\text{detect}} \in \{4, 6, 8, 10, 12, 16, 20, 24, 32\}$$

and

$$k_{\text{embed}} \in \{4, 6, 8, 10, 12, 16, 20, 24, 32\}.$$

Compare at least three placement choices: unaligned, guaranteed strips, and **optimized** centroid/landmark placement *if implemented*.

Evaluate connected-component count, partition accuracy (synthetic evaluation only), within-stress, trustworthiness, centroid/cross stress, clearance, running time and errors.

These are future experiment settings, not completed EXP-013 results.

Failure modes to report—not hide

1. **Overfragmentation:** sparse graph, small (k_{\rm detect}), (K) too large.
2. **False bridges:** manifold folds or nearby supports merge at large (k_{\rm detect}).
3. **Graph shortcut bias:** wrong graph geodesics even when partition is correct.
4. **Low-dimensional impossibility:** e.g., a torus surface cannot embed injectively in (\mathbb R^2) as a faithful continuous surface.
5. **Arbitrary alignment:** good internal geometry but huge cross/centroid stress.
6. **Evaluation leakage:** choosing (k) or (K) with generating labels, then claiming unsupervised recovery.
7. **Single-seed optimism:** cherry-picked 2D galleries do not prove robustness.

Comparison with known methods

Classical and established components of this pipeline:

- Isomap: Tenenbaum, de Silva & Langford (2000).
- Disconnected/multi-manifold approaches: D-C Isomap; M-Isomap; S-Isomap++.
- Neighborhood visualization: t-SNE; UMAP.
- Topological embedding controls: TopoMap and related persistence-aware methods.

Not an established novelty claim: independent kNN-Isomap alone.

Research target: a *specified and tested* separation-constrained placement objective that preserves each computed internal embedding while minimizing relative-placement distortion.

Three mathematical conclusions

1. Graph phase: connected components of symmetrized (k)-NN form a nested family as (k) increases, under deterministic neighbor ranking.

2. Spectral phase: classical MDS is obtained through
(\widehat B=-\tfrac12H\widehat D^{\circ2}H), then a positive rank-(d) spectral approximation.

3. Alignment phase: independent orthogonal transformations and translations preserve every computed intracomponent pair distance but alter cross-component placement.

Research question: how can we optimize the third phase with a positive hull clearance and a controlled ambient-distance cost?

Source and research status

Versioned run consulted

- EXP-010: `results/010-radius_density_and_rings/runs/20261019T061001Z-full/`.
- EXP-011: `results/011-isomap_baselines/runs/20261019T061001Z-full/`.
- Actual corresponding PNGs are embedded on the visual slides. EXP-011's `metadata.json` verifies **UMAP was installed**.
- The original EXP-011 code uses a radius or supplied-(K) MST partition, then rebuilds a 10-NN graph internally.
- **EXP-013 is proposed, not run or measured.**

The seminar should distinguish **proved mathematical identities**, **one-run empirical observations** and **proposed constrained optimization**.

Questions for the professor

1. Should detection use a single (k) or separate (k_{\rm detect}) and (k_{\rm embed})?
2. Which notion of **geometric separation** is appropriate: point clouds, convex hulls, or support estimates?
3. Should the placement objective preserve **centroid distances**, intercomponent landmarks, or all pairwise ambient distances?
4. What stronger result or benchmark would distinguish the project from prior multi-manifold Isomap methods?

$$\boxed{\text{graph recovery} \;\rightarrow\; \text{geodesic MDS} \;\rightarrow\; \text{separation–distortion optimization}}$$